

STATISTICAL COMPUTATION

Symbolic computation used to solve a Markov chain-based problem in statistical distribution theory

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Summary: Insufficient appreciation of the value of symbolic computation may restrict progress in statistical research and education. In this paper it is shown that complicated mathematical calculations can be eased considerably by applying appropriate computer algebraic techniques.

This is accomplished in the paper by algebraically solving a complex problem in statistical distribution theory. It is shown how the probability distribution and generating functions of a test statistic for a runs test for randomness can be found by symbolic computation, both under the null and the alternative hypothesis.

1. Introduction

Symbolic computation deals with mathematical computations on numbers, symbols, expressions and formulas as opposed to numeric computation that deals with numbers only. Many statistical packages have powerful numeric and graphic features but lack symbolic computing capabilities.

Mathematica® can be used to generate formulas, since the *Mathematica*®-based methods are symbolic. For example, Adamchik and Wagon (1996) showed how *Mathematica*® can be used to discover a formula which comes automatically with a proof.

The first part of the paper illustrates the advantages of symbolic computation by demonstrating elementary examples using a computer algebra system, namely *Mathematica*® (*Mathematica*® is a registered trademark of Wolfram Research Institute). In the second part of the paper it will be illustrated how *Mathematica*® can be used to provide a rich environment for mathematical manipulations in a complex statistical problem.

2. General discussion of symbolic computation

2.1 Advantages of symbolic computation

Development of statistical theory often demands complex mathematical calculations in which symbolic computations play an important role. There is a need for computational tools to assist the statistician in symbolic evaluation. Symbolic computing systems provide an environment for mathematical manipulations. These systems do nothing more complex than a statistician working by hand. They permit the statistician to concentrate on the structure of the calculation rather than on the detail of term-by-term evaluation. Symbolic computational systems may not always give simplified answers on symbolic solutions because some expressions are unsolvable or not expressible in terms of known forms. However, in many cases these systems can help with routine and tedious mathematical computations (Andrews and Stafford, 1993).

2.2 Examples illustrating the power of symbolic computation

In this section basic examples of the power of symbolic computation are presented using *Mathematica*®. Implementation is possible in other computer algebra languages that have similar basic features. The appearance of the statements closely resembles a segment of a *Mathematica*® session, which itself closely parallels the steps which would have been taken by manual calculation. The printed output is as the output appears on a printer. Input and output lines are identified by *In[]* and *Out[]* respectively.

As a first example, consider the case of power series expansion. Power series expansions will typically be addressed using the *Series* function in *Mathematica*®. The function *Series* can be viewed as a way of constructing Taylor series for functions. The standard formula for the Taylor series expansion of a convergent function $g(x)$ about the point $x = x_0$ is

$$g(x) = \sum_{k=0}^{\infty} g^{(k)}(x_0) \frac{(x - x_0)^k}{k!},$$

where $g^{(k)}(x)$ is the k -th derivative of $g(x)$.

The first five elements for the function e^x can easily be expressed using *Mathematica*® as follows (the order of the expansion is determined by the

Mathematica® notation, $\{x, 0, 4\}$). The function **Normal** turns the power series into an expression without the indication of higher order terms.

In[1]:= `Series[ex, {x, 0, 4}] // Simplify`

Out[1]= $1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + O[x]^5$

In[2]:= `Normal[%]`

Out[2]= $1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}$

The reverse problem can also be solved, as illustrated for the following infinite series:

In[3]:= $\sum_{n=0}^{\infty} x^n$

Out[3]= $\frac{1}{1-x}$

In *Mathematica*® the function *Binomial*[*n*, *m*] gives the binomial coefficient $\binom{n}{m} = \Gamma(n+1)/(\Gamma(m+1)\Gamma(n-m+1))$, where $\Gamma(n)$ is defined only for $n > 0$. The following example illustrates the use of the Binomial function in terms of a power series.

```

In[4]:= Sum[x^i Binomial[n, i], {i, 0, n}]
Out[4]:= (1+x)^n n! / Gamma[1+n]
In[5]:= FullSimplify[%]
Out[5]:= (1+x)^n

```

Note that *Mathematica*® often outputs a result in a form which the user would typically like to simplify, as illustrated in input statement number 5. The function *Collect*[*expr*, *x*], collects together terms involving the same powers of objects matching *x*. Consider the function $f(n) = \prod_{i=1}^n (1 + as^i)$. The terms involving *s* and *a* respectively can be collected together for $n = 3$ as follows:

```

In[6]:= f[n_] :=  $\prod_{i=1}^n (1 + a s^i)$ 

In[7]:= Collect[f[3], s]

Out[7]:=  $1 + a s + a s^2 + (a + a^2) s^3 + a^2 s^4 + a^2 s^5 + a^3 s^6$ 

In[8]:= Collect[f[3], a]

Out[8]:=  $1 + a^3 s^6 + a (s + s^2 + s^3) + a^2 (s^3 + s^4 + s^5)$ 

```

Examples of symbolic integration and differentiation using *Mathematica*® are illustrated below. Let the function $g[x]$ denote the probability density function of the normal(a,b) distribution. The definite integral over $(0, \infty)$ for this function is determined and the answer is given in *Out[12]*. The output gives an answer $\left(\frac{b \left(1 + \operatorname{Erf} \left[\frac{a}{\sqrt{2} \sqrt{b^2}} \right] \right)}{2\sqrt{b^2}} \right)$, provided the conditions $\operatorname{Re}[b^2] > 0$ and $\operatorname{Re}\left[\frac{a}{b^2}\right] < 0$ hold ($\operatorname{Re}[x] > 0$ means that the real part

of the complex number x is greater than zero). It also gives (in symbolic notation) the actual integral requested.

$$\begin{aligned}
 \text{In[9]} &:= \left[\ll \text{Statistics} \cdot \text{ContinuousDistributions} \cdot \right. \\
 \text{In[10]} &:= \left[g[x_] := \text{PDF}[\text{NormalDistribution}[a, b], x] \right. \\
 \text{In[11]} &:= \left[g[x] \right. \\
 \text{Out[11]} &:= \left[\frac{E^{-\frac{(a-x)^2}{2b^2}}}{b \sqrt{2\pi}} \right. \\
 \text{In[12]} &:= \left[\text{Integrate}[g[x], \{x, 0, \infty\}] \right. \\
 \text{Out[12]} &:= \left[\text{If} \left[\text{Re}[b^2] > 0 \ \&\& \ \text{Re} \left[\frac{a}{b^2} \right] < 0, \frac{b \left(1 + \text{Erf} \left[\frac{a}{\sqrt{2} \sqrt{b^2}} \right] \right)}{2 \sqrt{b^2}}, \int_0^\infty \frac{E^{-\frac{(a-x)^2}{2b^2}}}{b \sqrt{2\pi}} dx \right] \right.
 \end{aligned}$$

Next, the integral, $\int_0^\infty e^{-x^2} dx$ is determined symbolically in *Out[13]*.

$$\begin{aligned}
 \text{In[13]} &:= \left[\text{Integrate}[\text{Exp}[-x^2], \{x, 0, \text{Infinity}\}] \right. \\
 \text{Out[13]} &:= \left[\frac{\sqrt{\pi}}{2} \right.
 \end{aligned}$$

The treatment of symbolic differentiation is illustrated below in terms of a moment generating function. The solution for the fourth derivative of the $n(\mu, \sigma^2)$ moment generating function, $M(t) = e^{\mu t + \frac{1}{2} \sigma^2 t^2}$, is given in

Out[15]. Substituting t with 0 in Out[16] gives the fourth moment about the origin symbolically.

$$\text{In[14]:= } \left[M[t_]:=e^{\mu t+\frac{1}{2}\sigma^2 t^2} \right]$$

$$\text{In[15]:= } \left[M''''[t] \right]$$

$$\text{Out[15]= } \left[3\sigma^2 \left(e^{\mu t+\frac{\sigma^2 t^2}{2}} \sigma^2 + e^{\mu t+\frac{\sigma^2 t^2}{2}} (\mu + t\sigma^2)^2 \right) + (\mu + t\sigma^2) \left(2 e^{\mu t+\frac{\sigma^2 t^2}{2}} \sigma^2 (\mu + t\sigma^2) + (\mu + t\sigma^2) \left(e^{\mu t+\frac{\sigma^2 t^2}{2}} \sigma^2 + e^{\mu t+\frac{\sigma^2 t^2}{2}} (\mu + t\sigma^2)^2 \right) \right) \right]$$

$$\text{In[16]:= } \left[M''''[0] \right]$$

$$\text{Out[16]= } \left[3\sigma^2 (\mu^2 + \sigma^2) + \mu (2\mu\sigma^2 + \mu (\mu^2 + \sigma^2)) \right]$$

$$\text{In[17]:= } \left[M''''[0] - 4 M''''[0] M'[0] + 6 M''''[0] (M'[0])^2 - 3 (M'[0])^4 \right]$$

$$\text{Out[17]= } \left[-3\mu^4 + 6\mu^2 (\mu^2 + \sigma^2) + 3\sigma^2 (\mu^2 + \sigma^2) - 3\mu (2\mu\sigma^2 + \mu (\mu^2 + \sigma^2)) \right]$$

$$\text{In[18]:= } \left[\text{Simplify}[\%] \right]$$

$$\text{Out[18]= } \left[3\sigma^4 \right]$$

Many more interesting and exciting examples of the power of *Mathematica*® can be given. We will, however, now proceed to a complex problem which is solved symbolically with *Mathematica*®. This is the theme of the rest of the paper.

3. A Markov chain-based problem in Statistical Distribution Theory

3.1 The runs test: a test for randomness

A basic assumption in statistical inference is that the observed data constitute a random sample from some population. This assumption is often tested by **runs tests**, which are well-known nonparametric procedures (see for example Mood, 1940 and Gibbons, 1971). A run is defined to be a succession of one or more identical symbols, which are followed and preceded by a different symbol or no symbol at all in an ordered sequence of two or more types of symbols. Typically the data in a random sample is modelled as a Bernoulli process and the statistic used is the number of success runs in this process where the symbol under consideration is depicted as a success. The null hypothesis states that the symbols follow one another in a random way, that is, there is no tendency among the symbols to exhibit a definite pattern. This implies that the number of runs should neither be too small nor too large. The alternative hypothesis is often simply nonrandomness, which can be manifested in different ways such as cycles (many runs), monotone increasing/decreasing series of observations (few runs) etc. (see Smit, 1996 for an investigation of different alternatives to randomness).

3.2 Distribution of $G_{n,k}$

The results in Fu and Koutras (1994) present a simple and unified approach for the distribution theory of runs tests based on a finite Markov chain imbedding technique. Fu and Koutras considered statistics based on success runs of both identically and nonidentically distributed independent Bernoulli trials. One important and frequently used statistic of success runs associated with independent Bernoulli trials $X_1, \dots, X_n$, success probabilities p_t and failure probabilities $q_t = 1 - p_t$, for $t = 1, \dots, n$ is defined by:

$G_{n,k}$: number of success runs of size greater than or equal to k (k fixed, $1 \leq k \leq n$).

The index t indicates that the Bernoulli variables are not necessarily identically distributed. Consider the following example: Two persons A and B play a game n times. Each successive time A plays the game, his chances to win could increase due to learning and experience. If S denotes a win for A and F a win for B, typical data could be FSFSSFSSSSFS. In this case $n = 12$, $k = 1$ and $G_{12,1} = 4$ could be used to test whether the series is random or whether A has indeed improved his chances to win.

The importance of the paper by Fu and Koutras (1994) for the problem of randomness lies in the theoretical results which they provide and which can be used to find the exact size and power of runs tests.

Fu and Koutras' fundamental idea of finding the exact distribution of a runs statistic is to transform the way the success runs are counted into a finite Markov chain with a proper state space and transition probability matrix. In order to better understand the symbolic calculations we use, a brief discussion of terminology and notation is given in the next section.

3.3 Definitions and terminology

Let Λ_t be the $m \times m$ transition probability matrix of the finite Markov chain $(\{Y_t : t \in \Gamma_n\}, \Omega)$, with $\Gamma_n = \{0, 1, \dots, n\}$ an index set and $\Omega = \{a_1, \dots, a_m\}$ a finite state space. For finding the distribution of $G_{n,k}$ the state space is defined by

$$\Omega(G_{n,k}) = \{(x, i) : x = 0, 1, \dots, \ell \text{ and } i = -1, 0, \dots, k-1\} - \{(0, -1)\}, \ell = \left\lfloor \frac{n+1}{k+1} \right\rfloor.$$

The Markov chain $\{Y_t\}$ is expressed in terms of the states (x, i) , where x denotes the number of success runs and i is related to the length of the current success run at observation t . The state space $\Omega(G_{n,k})$ is partitioned in terms of elements C_0 and C_x as follows:

$$C_0 = \{(0, i) : i = 0, 1, \dots, k-1\} \text{ and}$$

$$C_x = \{(x, i) : i = -1, 0, 1, \dots, k-1\}, \quad x = 1, 2, \dots, \ell.$$

Let U_r be a $1 \times m$ unit vector having 1 at the r -th coordinate and 0 elsewhere, and let U'_r be the transpose of U_r . The $1 \times m$ vector $U(C_x)$ is defined (for every C_x) as:

$$U(C_x) = \sum_{r: a_r \in C_x} U_r.$$

Given the framework and notation above, it follows that $P(G_{n,k} = x) = P(Y_n \in C_x)$. From this, one sees that (Fu and Koutras)

$$P(G_{n,k} = x) = \pi_0 \left(\prod_{t=1}^n \Lambda_t \right) U'(C_x), \quad (3.3.1)$$

where the initial probability of the Markov chain is given by $\pi_0 = (P(Y_0 = a_1), P(Y_0 = a_2), \dots, P(Y_0 = a_m))$ and Λ_t is a transition probability matrix.

In general Λ_t is a bidiagonal blocked matrix of the following form:

$$\Lambda_t = \left[\begin{array}{ccccccc} A_t & p_t e'_k & & & & & \\ & p_t & q_t e_1 & & & & \\ \hline & & & A_t & p_t e'_k & & \\ & & & & p_t & q_t e_1 & \\ \hline & & & & & & \ddots \\ & & & & & & p_t & q_t e_1 \\ \hline & & & & & & & A_t^* \end{array} \right]$$

where $e_1 = (1, 0, \dots, 0)$ and $e_k = (0, \dots, 0, 1)$ are unit $1 \times k$ vectors and the blank positions represent zeros. The dimension (m) of Λ_t is given by $m = (\ell + 1)(k + 1) - 1$.

We will refer to the parts separated by dotted lines as rows of Λ_t in the development of the calculation algorithm.

In the above expression A_t is a $k \times k$ matrix of the form:

$$A_t = \left[\begin{array}{ccccccc} q_t & p_t & 0 & 0 & \cdot & \cdot & \cdot \\ q_t & 0 & p_t & 0 & \cdot & \cdot & \cdot \\ q_t & 0 & 0 & p_t & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ q_t & 0 & 0 & 0 & \cdot & \cdot & \cdot \end{array} \right]$$

A_t^* is a matrix of the same form as A_t except for its last row, which is replaced by $(0, 0, \dots, 0, 1)$.

This result (3.3.1) of Fu and Koutras helps one to find the exact distribution of $G_{n,k}$ both under the null hypothesis (randomness) and the alternative. It also enables one to find expressions for the moments and probability generating functions of statistics like $G_{n,k}$.

Our purpose will be to develop a *Mathematica*® notebook for finding symbolic expressions for $P(G_{n,k} = x)$, the associated moments and the probability generating function.

3.4 Symbolic calculation of the transition probability matrix Λ_t

Our first goal was to develop an algorithm to calculate the series of transition probability matrices Λ_t , $t = 1, \dots, n$ for a given n and k . The *Mathematica*® code was developed for the distribution of $G_{n,k}$ in terms of general n , k and p_t . Since larger choices of the parameters n and k will result in enormous outputs, we only discuss the code for the special case where $n = 5$, $k = 2$ and $p_t = c_t + \delta$, $t = 1, \dots, n$ as shown below (note that c_t is a function of t only and $0 < c_t + \delta < 1$). The success probability p_t is defined by input statement number 4 (or *In*[4]).

```
In[4]:= | p[t_, δ_] := c[t] + δ |
```

The matrix Λ_t is divided into different parts namely A_t , e_k , and A_t^* . For a given n , k and p_t , these parts can be given symbolically as shown in the output statements below. The user can specify the value of p_t at the beginning of the notebook and values for n and k at any stage. The computations can be kept in symbolic format (as is illustrated in the outputs) until the end of the notebook, where given values for n , k and p_t can be defined to determine the required expressions (for example, the moment generating function of $G_{n,k}$). In other words, when the notebook is to be used for a specific application, one can omit those input statements given for illustrative purposes only and define the user specific values for the parameters n , k and p_t . In the *Mathematica*®

code to follow *Out[8]*, *Out[10]* and *Out[12]* give A_t , e_t and A_t^* respectively (for $n = 5$, $k = 2$).

```
In[8]:= a[t, 2, 4] // MatrixForm
Out[8]//MatrixForm=

$$\begin{pmatrix} 1-\delta-c[t] & \delta+c[t] \\ 1-\delta-c[t] & 0 \end{pmatrix}$$

In[10]:= e[2, 2] // MatrixForm
Out[10]//MatrixForm=

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

In[12]:= astar[t, 2, 4] // MatrixForm
Out[12]//MatrixForm=

$$\begin{pmatrix} 1-\delta-c[t] & \delta+c[t] \\ 0 & 1 \end{pmatrix}$$

```

From the output of the matrix A_t^* , it is clear that its form is the same as that of A_t except for its last row.

The next steps construct the first two rows of the transition matrix Λ , given in Section 3.3 (part above first dotted line). This is established by joining A_t and $p_t e'_k$, for row 1, and by joining p_t and $q_t e_1$ for row 2.

```

In[14]:= blk[t, 5, 2, d] // MatrixForm
Out[14]/MatrixForm=

$$\begin{bmatrix} 1-\delta-c[t] & \delta+c[t] & 0 & 0 & 0 & 0 & 0 & 0 \\ 1-\delta-c[t] & 0 & \delta+c[t] & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$


In[18]:= blkkr[t, 5, 2, d] // MatrixForm
Out[18]/MatrixForm=

$$\begin{bmatrix} 1-\delta-c[t] & \delta+c[t] & 0 & 0 & 0 & 0 & 0 & 0 \\ 1-\delta-c[t] & 0 & \delta+c[t] & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \delta+c[t] & 1-\delta-c[t] & 0 & 0 & 0 & 0 \end{bmatrix}$$


```

Out[18] gives $\begin{pmatrix} A_t & p_t e'_k \\ p_t & q_t e_1 \end{pmatrix}$

The middle rows of the matrix (separated by the first and last dotted lines) are found by repeatedly shifting each of the blocks A_t , $p_t e'_k$ and $(p_t, q_t e_1)$ $k+1$ units to the right until $\ell-1$ such shifts have been done.

```

In[20]:= Mid[t, 5, 2, δ] // MatrixForm
Out[20]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 1-\delta-c[t] & \delta+c[t] & 0 & 0 & 0 \\ 0 & 0 & 0 & 1-\delta-c[t] & 0 & \delta+c[t] & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \delta+c[t] & 1-\delta-c[t] & 0 \end{pmatrix}$$


```

The last row below the last dotted line of Λ_t consists of a $k \times m$ table with $m - k$ zero columns and A_k^* in the last k columns.

```

In[22]:= Lr[t, 5, 2, δ] // MatrixForm
Out[22]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1-\delta-c[t] & \delta+c[t] \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$


```

By joining the different parts together, the final transition probability matrix, Λ_t , can be given symbolically.

```

In[24]:= Lambda[t, 5, 2, δ] // MatrixForm
Out[24]//MatrixForm=

$$\begin{pmatrix} 1-\delta-c[t] & \delta+c[t] & 0 & 0 & 0 & 0 & 0 & 0 \\ 1-\delta-c[t] & 0 & \delta+c[t] & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \delta+c[t] & 1-\delta-c[t] & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1-\delta-c[t] & \delta+c[t] & 0 & 0 & 0 \\ 0 & 0 & 0 & 1-\delta-c[t] & 0 & \delta+c[t] & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \delta+c[t] & 1-\delta-c[t] & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1-\delta-c[t] & \delta+c[t] \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$


```

Since the code is general, other choices for n , k and p_t can also be made.

4. Probability function, moments and probability generating function of $G_{n,k}$

4.1 Probability function

In order to construct the probability function of $G_{n,k}$, namely, $P(G_{n,k} = x) = \pi_0 \left(\prod_{t=1}^n \Lambda_t \right) U'(C_x)$, the function $U'(C_x) = \sum_{r: a_r \in C_x} U'_r$ must first be determined. This is easily done in *Mathematica*®.

For calculation of the probability function of $G_{n,k}$, all the different components are multiplied to form the probability density function (denoted by $\text{prob}[n, k, \delta, x]$). An example for $P(G_{n,k} = 1)$, $p_t = c_t + \delta$, $n = 5$ and $k = 2$ is given symbolically in *Out[38]*.

```
In[38]:= prob[5, 2, δ, 1]
```

```
Out[38]:= ((δ+c[1.]) (δ+c[2.]) (1.-1. δ-1. c[3.]) (1.-1. δ-1. c[4.]) +
  ((1.-1. δ-1. c[1.]) (δ+c[2.]) (δ+c[3.]) + (δ+c[1.]) (δ+c[2.]) (δ+c[3.]))
  (1.-1. δ-1. c[4.]))
  (1.-1. δ-1. c[5.]) +
  (δ+c[1.]) (δ+c[2.]) (1.-1. δ-1. c[3.]) (δ+c[4.]) (1.-1. δ-1. c[5.]) +
  (((1.-1. δ-1. c[1.]) (1.-1. δ-1. c[2.]) + (δ+c[1.]) (1.-1. δ-1. c[2.]))
  (δ+c[3.]) (δ+c[4.]) + ((1.-1. δ-1. c[1.]) (δ+c[2.]) (δ+c[3.]) +
  (δ+c[1.]) (δ+c[2.]) (δ+c[3.])) (δ+c[4.]))
  (1.-1. δ-1. c[5.]) +
  ((δ+c[1.]) (δ+c[2.]) (1.-1. δ-1. c[3.]) (1.-1. δ-1. c[4.]) +
  ((1.-1. δ-1. c[1.]) (δ+c[2.]) (δ+c[3.]) + (δ+c[1.]) (δ+c[2.]) (δ+c[3.]))
  (1.-1. δ-1. c[4.]))
  (δ+c[5.]) +
  (((1.-1. δ-1. c[1.]) (1.-1. δ-1. c[2.]) + (δ+c[1.]) (1.-1. δ-1. c[2.]))
  (1.-1. δ-1. c[3.]) +
  (1.-1. δ-1. c[1.]) (δ+c[2.]) (1.-1. δ-1. c[3.])) (δ+c[4.]) (δ+c[5.]) +
  (((1.-1. δ-1. c[1.]) (1.-1. δ-1. c[2.]) + (δ+c[1.]) (1.-1. δ-1. c[2.]))
  (δ+c[3.]) (δ+c[4.]) + ((1.-1. δ-1. c[1.]) (δ+c[2.]) (δ+c[3.]) +
  (δ+c[1.]) (δ+c[2.]) (δ+c[3.])) (δ+c[4.]))
  (δ+c[5.]
```

The complete probability density function of $G_{n,k}$ is given partially (first few terms in each case) for $c[t] = \frac{1}{t+1}$ in *Out[41]*, since very long symbolic expressions are generated.

```

In[40]:= c[t_] := 1/(t + 1)

In[41]:= probTable[5, 2, 4]

Out[41]//TableForm=

```

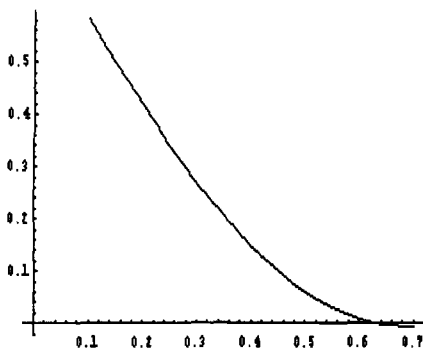
$P(G_{5,2}=0)$	$= (0.833333 - 1. \delta) (0.2 + \delta) ((0.5 - 1. \delta) (0.75 - 1. \delta) (0.333333 + \delta) + (0.75 - 1. \delta) (0.166667 + \delta) (0.5 + \delta))$
$P(G_{5,2}=1)$	$= (0.75 - 1. \delta) (0.833333 - 1. \delta) (0.2 + \delta) (0.333333 + \delta) (0.5 + \delta) + (0.166667 - 1. \delta) (0.5 + \delta) (0.2 + \delta)$
$P(G_{5,2}=2)$	$= (0.75 - 1. \delta) (0.166667 + \delta) (0.2 + \delta) (0.333333 + \delta) (0.5 + \delta)$

A symbolic expression and a plot of the probabilities, $P(G_{5,2} = 0)$ for δ ranging from 0.1 to 0.7 (on the horizontal axis) is given in *Out[42]* and *Out[43]* respectively.

```
In[42]:= prob[5, 2, δ, 0] // FullSimplify
```

```
Out[42]:= -1. (-0.958319 + δ) (-0.792992 + δ) (-0.639509 + δ) (0.713276 + δ)
          (2.12754 + δ)
```

```
In[43]:= graf0 = Plot[-1. (-0.958319200853715713 + δ)
                    (-0.792991736538151847 + δ) (-0.639509189974692393 + δ)
                    (0.713276403000456007 + δ) (2.12754364436618976 + δ),
                    {δ, 0.1, 0.7} ]
```



```
Out[43]:= - Graphics -
```

4.2 Moments and probability generating function

The moments of $G_{n,k}$, $E(G_{n,k})^r$, $r = 1, 2, \dots$, can be written as $E(G_{n,k})^r = \pi_0 \left(\prod_{t=1}^n \Lambda_t \right) V_r'$, where $V_r' = \sum_{x=0}^{\ell} x^r U'(C_x)$ (see Fu and Koutras, 1994).

Similarly, the probability-generating function of $G_{n,k}$ can be written as $\varphi_{n,k}(s) = \pi_0 \left(\prod_{t=1}^n \Lambda_t \right) W'(s)$, where $W'(s) = \sum_{x=0}^{\ell} s^x U'(C_x)$.

The first four moments ($EGnk[n,k,\delta]$) and probability-generating function (denoted by $\text{phi}[s, n, k, \delta]$) can easily be constructed with *Mathematica*®, as shown below. At this point the previously used choices for n , k and p_t , namely $n = 5$, $k = 2$ and, $p_t = \frac{1}{1+t} + \delta$, with $\delta = 0$, are taken since enormous outputs will be generated in the general case. Note again that the user can specify the values of the parameters for his specific case, by adjusting the relevant *Mathematica*® code.

```

In[51]:= EGnk[5, 2, 0]
Out[51]= {0.266667, 0.275, 0.291667, 0.325}

In[54]:= phi[s, 5, 2, 0]
Out[54]= 0.7375 + 0.258333 s + 0.00416667 s^2

```

The moment generating function (given in $\text{ln}[55]$) can be calculated from $M_G(t) = \varphi_{n,k}(e^t)$ and developed as a series to find the first four moments in *Mathematica*® as shown below. The function **Collect** sorts the coefficients of t^r , $r = 1, 2, \dots$ in the proper order.

```

In[55]:= phi[Normal[Series[Exp[t], {t, 0, 4}]], 5, 2, 0]

Out[55]:= 0.7375 + 0.258333 (1. + t + 0.5 t^2 + 0.166667 t^3 + 0.0416667 t^4) +
          0.00416667 (1. + t + 0.5 t^2 + 0.166667 t^3 + 0.0416667 t^4)^2

In[56]:= Collect[%, t]

Out[56]:= 1. + 0.266667 t + 0.1375 t^2 + 0.0486111 t^3 + 0.0135417 t^4 +
          0.00104167 t^5 + 0.000289352 t^6 + 0.0000578704 t^7 + 7.2338 x 10^-6 t^8

```

Multiplying the coefficients of t^r , $r = 1, 2, \dots$, by $r!$ one gets the moments of $G_{n,k}$ (compare *Out[51]* and *Out[56]*).

5. Conclusion

The importance of symbolic computation on computer for statistical research was illustrated for a few simple and one complex problem. Specifically, a *Mathematica*® notebook for finding the probability function, moments and some generating functions for a commonly used runs test statistic was developed under a general hypothesis. The symbolic notation enables one to study, in general terms, important properties of such a test, such as moments of the test statistic, the test's power, et cetera.

The symbolic notation in which computer algebra systems can give results is an advantage over some other statistical software packages. It may be

worthwhile to give serious attention to the value of such systems in order to assist in finding solutions in algebraic form of many statistical distribution theory problems.

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